

TARUN REDDY

AI/ML Engineer | Gen AI

Mail: tarunreddym11@gmail.com | Ph no: +1(351) 220-1157 | LinkedIn: www.linkedin.com/in/tarun-reddy-952a77359

PROFESSIONAL SUMMARY

- Over 4+ years of experience as an AI/ML Engineer and Python Developer, specializing in Machine Learning, Generative AI, LLMs, NLP, MLOps, and AI solutions.
- Proficient in Python, SQL, Java, C++, Bash, and TypeScript for AI applications, APIs, automation, and data processing.
- Experienced in Machine Learning using Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Predictive Modeling, Anomaly Detection, Scikit-learn, XGBoost, and LightGBM.
- Skilled in Deep Learning with PyTorch, TensorFlow, Keras, Neural Networks, CNN, RNN, LSTM, GRU, Transformers, Attention Mechanisms, and Transfer Learning.
- Experienced in Generative AI and LLMs using GPT, Claude, Llama, Gemini, Mistral, Hugging Face, OpenAI API, Azure OpenAI, Amazon Bedrock, Prompt Engineering, and Function Calling.
- Experienced in RAG application development using RAG, Embeddings, Document Chunking, Re-Ranking, Semantic Search, Hybrid Search, Knowledge Bases, Pinecone, FAISS, ChromaDB, Qdrant, Milvus, and pgvector.
- Proficient in Agentic AI and AI Agents using LangChain, LangGraph, LlamaIndex, Semantic Kernel, CrewAI, AutoGen, OpenAI Agents SDK, Tool Calling, Agent Memory, Agent Orchestration, and Multi-Agent Systems.
- Hands-on experience with LLM Fine-Tuning and Inference using SFT, PEFT, LoRA, QLoRA, DPO, RLHF, Model Distillation, Model Quantization, vLLM, TensorRT-LLM, NVIDIA Triton, and ONNX Runtime.
- Knowledge of AI Evaluation and Security including LLM Evaluation, RAG Evaluation, Hallucination Detection, RAGAS, DeepEval, AI Guardrails, Prompt Injection Detection, and PII Protection.
- Experienced in MLOps and LLMOps using ML Pipelines, Experiment Tracking, Model Registry, Model Versioning, Model Validation, Model Deployment, Model Monitoring, MLflow, Kubeflow, Weights & Biases, DVC, and Airflow.
- Proficient in AWS, Azure, and GCP AI platforms, including SageMaker, Bedrock, Azure Machine Learning, Azure OpenAI, Azure AI Foundry, Azure AI Search, Vertex AI, EKS, AKS, and GKE.
- Experienced in AI Infrastructure & Production Engineering using Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins, Azure DevOps, CUDA, GPU Computing, FastAPI, REST APIs, Microservices, & AI Observability.
- Strong communication, analytical, problem-solving, collaboration, adaptability, and ownership skills, with experience delivering scalable, reliable, secure, and production-ready AI solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, C++, Bash, TypeScript
- **Machine Learning:** Machine Learning, Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Predictive Modeling, Scikit-learn, XGBoost, LightGBM
- **Deep Learning:** PyTorch, TensorFlow, Keras, Neural Networks, CNN, RNN, LSTM, GRU, Transformers, Transfer Learning
- **Generative AI & LLMs:** Generative AI, LLMs, Foundation Models, GPT, Claude, Llama, Gemini, Mistral, Hugging Face, OpenAI API, Azure OpenAI, Amazon Bedrock, Prompt Engineering, Function Calling
- **RAG & Vector Databases:** RAG, Embeddings, Semantic Search, Hybrid Search, Vector Search, Document Chunking, Re-Ranking, Knowledge Bases, Pinecone, FAISS, ChromaDB, Qdrant, Milvus, pgvector
- **AI Agents & Frameworks:** Agentic AI, AI Agents, Multi-Agent Systems, Agent Orchestration, Tool Calling, LangChain, LangGraph, LlamaIndex, Semantic Kernel, CrewAI, AutoGen, OpenAI Agents SDK
- **LLM Fine-Tuning & Inference:** Fine-Tuning, Instruction Tuning, SFT, PEFT, LoRA, QLoRA, DPO, RLHF, Model Quantization, vLLM, TensorRT-LLM, NVIDIA Triton, ONNX Runtime, Model Serving
- **AI Evaluation & Safety:** LLM Evaluation, RAG Evaluation, Model Evaluation, Prompt Evaluation, Hallucination Detection, RAGAS, DeepEval, TruLens, AI Guardrails, Prompt Injection Detection, PII Protection, Responsible AI
- **MLOps & ML Engineering:** MLOps, LLMOps, ML Pipelines, Experiment Tracking, Model Registry, Model Deployment, Model Monitoring, MLflow, Kubeflow, Weights & Biases, DVC, Airflow
- **Cloud & AI Platforms:** AWS, Azure, GCP, SageMaker, Amazon Bedrock, Azure Machine Learning, Azure OpenAI, Azure AI Foundry, Azure AI Search, Vertex AI, Amazon S3, AWS Lambda, EKS, AKS, GKE
- **Data & DevOps:** ETL, ELT, Data Pipelines, Apache Spark, PySpark, Apache Kafka, Databricks, Delta Lake, Snowflake, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps
- **AI Architecture & Production:** AI System Design, ML System Design, RAG Architecture, LLM Architecture, AI Agents, Microservices, REST APIs, FastAPI, CI/CD, Infrastructure as Code, GPU Computing, Distributed Training, Distributed Inference, AI Observability, Performance Optimization, AI Security, Production AI

PROFESSIONAL WORK EXPERIENCE

Johnson & Johnson | AI/ML Engineer | Generative AI

Oct 2025 – Present

Responsibilities

- Architected and developed Generative AI and Machine Learning applications using Python, SQL, LLMs, GPT, Claude, Llama, Gemini, Mistral, Hugging Face, OpenAI API, Azure OpenAI, and Amazon Bedrock for pharmaceutical workflows.

- Developed RAG solutions using Advanced RAG, document chunking, embeddings, semantic search, hybrid search, re-ranking, knowledge bases, and vector search to support enterprise knowledge and information retrieval.
- Implemented vector database solutions using Pinecone, FAISS, Qdrant, ChromaDB, Milvus, and pgvector to improve context retrieval and LLM response generation.
- Built AI agents and agentic workflows using LangChain, LangGraph, LlamaIndex, and OpenAI Agents SDK, with tool calling, function calling, agent memory, workflow orchestration, and multi-agent systems.
- Developed scalable AI services using Python, FastAPI, REST APIs, and Microservices, integrating LLM inference, RAG components, backend services, and enterprise applications.
- Implemented LLM evaluation and AI safety using RAGAS, DeepEval, TruLens, response validation, hallucination detection, groundedness, faithfulness, AI guardrails, prompt injection detection, PII protection, and Responsible AI.
- Supported MLOps and LLMops using MLflow, experiment tracking, model versioning, model validation, model deployment, model monitoring, AI observability, and automated model lifecycle management.
- Containerized and deployed AI applications using Docker, Kubernetes, Helm, AWS, Azure, CI/CD, and cloud-native deployment practices for scalable development, testing, and production environments.
- Optimized LLM inference and model serving using NVIDIA CUDA, GPU Computing, model quantization, ONNX Runtime, vLLM, TensorRT-LLM, and NVIDIA Triton for efficient real-time AI applications.
- Automated CI/CD and Infrastructure as Code using GitHub Actions, Jenkins, Terraform, and Argo CD for AI deployments.

Environment: Python, SQL, Machine Learning, Generative AI, LLMs, GPT, Claude, Llama, Gemini, Mistral, Hugging Face, OpenAI API, Anthropic API, Azure OpenAI, Amazon Bedrock, Prompt Engineering, Prompt Optimization, RAG, Advanced RAG, Embeddings, Semantic Search, Hybrid Search, Re-Ranking, Pinecone, FAISS, Qdrant, ChromaDB, Milvus, pgvector, LangChain, LangGraph, LlamaIndex, OpenAI Agents SDK, AI Agents, Agentic AI, Multi-Agent Systems, FastAPI, REST APIs, Microservices, MLflow, RAGAS, DeepEval, TruLens, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Jenkins, Argo CD, NVIDIA CUDA, GPU Computing, vLLM, TensorRT-LLM, NVIDIA Triton, ONNX Runtime, AWS, Azure, MLOps, LLMops, CI/CD, AI Observability, Production AI.

JPMorgan Chase & Co. | AI/ML Engineer - GenAI | NLP | MLOps

Dec 2024 – Oct 2025

Responsibilities

- Developed Machine Learning and NLP solutions using Python, Scikit-learn, XGBoost, LightGBM, classification, regression, clustering, anomaly detection, and predictive modeling for financial applications.
- Built NLP pipelines using Transformers, PyTorch, text preprocessing, Named Entity Recognition, text classification, and information extraction for document processing.
- Processed large financial datasets using Apache Spark, PySpark, Databricks, Delta Lake, Snowflake, and Apache Kafka, supporting structured and unstructured data workflows.
- Implemented LLM fine-tuning using SFT, PEFT, LoRA, and QLoRA for domain-specific financial applications.
- Built GenAI solutions using OpenAI, Azure OpenAI, Bedrock, Hugging Face, and LangChain for financial applications.
- Improved model development efficiency by 25% through MLflow, Weights & Biases, experiment tracking, model validation, model registries, and model versioning.
- Built FastAPI and REST API services using Docker and Kubernetes for scalable model serving and application integration.
- Reduced deployment time by 30% by automating GitLab CI/CD, Jenkins, Azure DevOps, Terraform, and Kubernetes workflows for production MLOps.

Environment: Python, SQL, Java, C++, Machine Learning, NLP, Scikit-learn, XGBoost, LightGBM, Transformers, PyTorch, TensorFlow, GPT, OpenAI API, Azure OpenAI, Amazon Bedrock, Hugging Face, LangChain, Semantic Kernel, Instruction Tuning, SFT, PEFT, LoRA, QLoRA, Apache Spark, PySpark, Kafka, Databricks, Delta Lake, Snowflake, PostgreSQL, MLflow, Weights & Biases, FastAPI, Docker, Kubernetes, GitLab CI/CD, Jenkins, Azure DevOps, Terraform, AWS, Azure, MLOps, LLMops

Fortis Healthcare | Python Developer | India

Jan 2022 – Jul 2024

Responsibilities

- Built Python-based backend services using SQL, FastAPI, REST APIs, Bash, & Microservices to support healthcare workflows.
- Implemented data processing workflows using Python and SQL to extract, transform, clean, and validate data.
- Developed database solutions using PostgreSQL, MongoDB, & Redis for data storage, caching, & application performance.
- Processed large datasets using Apache Spark and PySpark to support analytics and data preparation.
- Improved application reliability through automated testing, logging, exception handling, debugging, and monitoring.
- Integrated internal and external systems using REST APIs and JSON, working with development, QA, and business teams.
- Containerized applications using Docker and managed source code with Git in Linux environments.
- Supported Agile development, unit testing, API integration, backend development, and application monitoring.

Environment: Python, SQL, Bash, FastAPI, REST APIs, Microservices, PostgreSQL, MongoDB, Redis, Apache Spark, PySpark, ETL, Data Processing, Data Validation, Docker, Git, Linux, JSON, Unit Testing, Logging, Backend Development, Agile

EDUCATION

- **Master's degree in Computer Science** from Clark University, Massachusetts, United States.
- **Bachelor of Technology in Computer Science** from Jawaharlal Nehru Technological University Hyderabad (JNTUH), India.